## Process structure

10 candidate structures

Chilling  Forcing

Forcing only

2 structures

**Sequential**

4 structures

Parallel

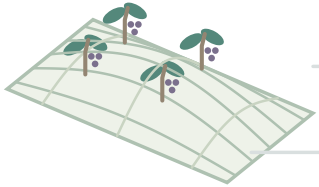
4 structures

**Selected: CB-FL-S**

Beta chilling → linear forcing

## Further improvement

### Terrain-adjusted temperature



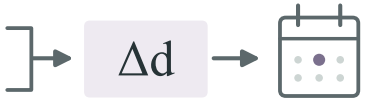
Elevation offset

Recalibration

### Process-guided residual learning

Process states

Environment




Grouped out-of-fold errors

Both corrections evaluated across all 10

## Grapevine phenology

Regional test MAE (days)

